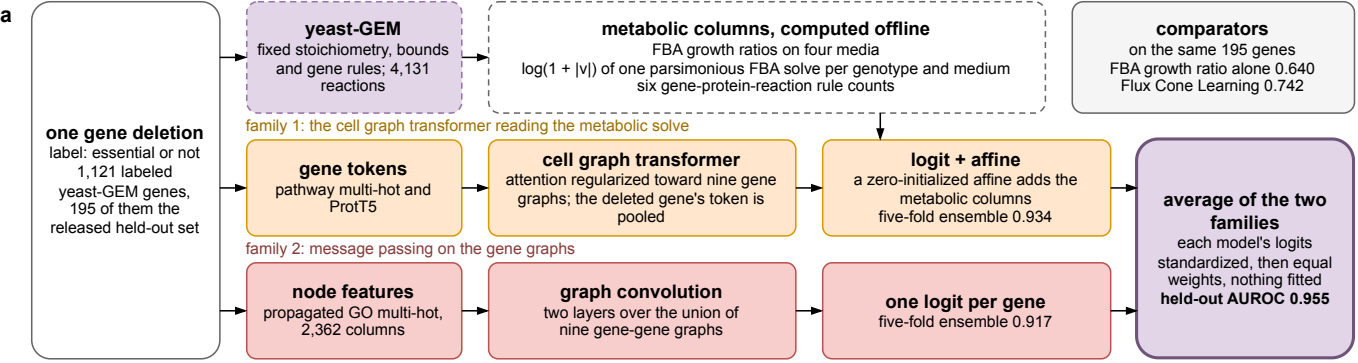
a
one gene deletion
label: essential or not
1,121 labeled
yeast-GEM genes,
195 of them the
released held-out set
yeast-GEM
fixed stoichiometry, bounds
and gene rules; 4,131
reactions
metabolic columns, computed offline
FBA growth ratios on four media
log(1 + |v|) of one parsimonious FBA solve per genotype and medium
six gene-protein-reaction rule counts
comparators
on the same 195 genes
FBA growth ratio alone 0.640
Flux Cone Learning 0.742
family 1: the cell graph transformer reading the metabolic solve
gene tokens
pathway multi-hot and
ProtT5
cell graph transformer
attention regularized toward nine gene
graphs; the deleted gene's token is
pooled
logit + affine
a zero-initialized affine adds the
metabolic columns
five-fold ensemble 0.934
family 2: message passing on the gene graphs
node features
propagated GO multi-hot,
2,362 columns
graph convolution
two layers over the union of
nine gene-gene graphs
one logit per gene
five-fold ensemble 0.917
average of the two
families
each model's logits
standardized, then equal
weights, nothing fitted
held-out AUROC 0.955